Student Answer Script View

058-END SEMSTER-ODD-2025 Answer Sheet

Course:	Computer Science and Engineering
Year/Sem:	Semester 7
Subject Name:	Cloud Computing
Exam Date:	25-Nov-2025

33.00

50.00

Comments submitted by student.

Q.No : 1 A) **Score : 3.50 / 5.00**

A large university, GlobalTech University, plans to implement a cloud-based Learning Management System (LMS) for students and faculty. The system will store student records, assignments, and research data. The university wants to:

- Ensure data security and privacy for sensitive academic information,
- Allow scalability to handle thousands of users during exams or admissions,
- Facilitate collaboration with other universities for joint research projects, and
- Keep operational costs under control without maintaining too much hardware.

As a Cloud Computing Consultant, recommend the most appropriate cloud deployment model (Public, Private, Community, or Hybrid) for *GlobalTech University*.

Apply the features of the chosen model to justify how it meets the university's requirements.

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Solution:- For GlobalTech University as a cloud consultant I will choose the community cloud. I choose this cloud model because it meets the requirements as listed in the requirement section.

1. Ensure data security for sensitive Academic information.

Since, community cloud often has features like identification, authentication and authorized access, we can ensure security for sensitive academic information.

2. Allow scalability to handle thousands of users during exams or admission.

i) Once we register a user in the community we can ensure, the users that are part of our GlobalTech University cloud, we can handle thousands of users, whether it is exams of admission test.

3. Facilitate collaboration with other universities for joint research projects

In community clouds, we can also partner community different university clouds to gain access and authorization for joint research projects

Keep operational cost under control without maintaing too much hard ware.

i) since in community clouds we use the resources of the joint shared community, infrastructure and resource costs are kept undercontrol without maintaining too much hardware.

Q.No : 1 B)	**Score : 4.00 / 5.00**

A small digital marketing company, *AdSphere Media*, is planning to launch a new web application that provides real-time campaign analytics for its clients. The company's goals are to:

- Develop and deploy the application quickly without managing complex infrastructure,
- Use ready-made software tools for email marketing and data visualization,
- Scale resources automatically as the number of users increases, and
- Keep overall IT maintenance and cost low.

As a Cloud Computing Consultant, suggest the most suitable cloud service model(s) (*IaaS, PaaS, or SaaS*) for *AdSphere Media.*

Apply the features of the chosen model(s) to justify how they address the company's needs.

Solution:- If I was a cloud consultant, I would use PaaS (Platform as a Service) for the marketing company Adsphere Media, whose aim to provide real-time campaign analytics for its clients. I would choose this because of the following reasons:-

i) Develop and deploy the application quickly without managing complex infrastructure. :

Here our aim for the company Adsphere media is to provide campaign analytics, we would not need complex infrastructure, as we are not providing services on them, we would use some one elses infrastructure, for a fast application deployment

Use ready-made software tools for email marketing and data-visualization. :

Our business does not focus on the Building new software services, So we will use other software services like yahoo and hotmail, gmail emailing services for marketing and Power BI (software service) for data visualization.

Scale resources automatically as number of users increases. :

we can quickly scale resources as our User base increases to avoid the cost aspect condition of our adsphere media.

ii) As we scale, and grow in revenue, we can slowly adopt independent services and infrastructure to reduce cost.

Q.No : 2 B) **Score : 0.00 / 5.00**

In a hybrid cloud, workloads are inconsistently balanced across VMs.

a) Analyze the resource allocation metrics that lead to imbalance.

b) Evaluate how dynamic VM migration can improve overall load distribution.

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Q.No : 3 A)	**Score : 5.00 / 5.00**

A smart city uses a Message Bus with a Publish-Subscribe model to deliver traffic and pollution alerts to citizen apps.

a) Analyze how does the message bus reduce direct dependencies between services in this system?

b) Which filtering method (topic-based or content-based) would you apply for sending only pollution alerts to subscribed users? Justify.

c) Explain how event-driven alerts help city authorities take immediate action.

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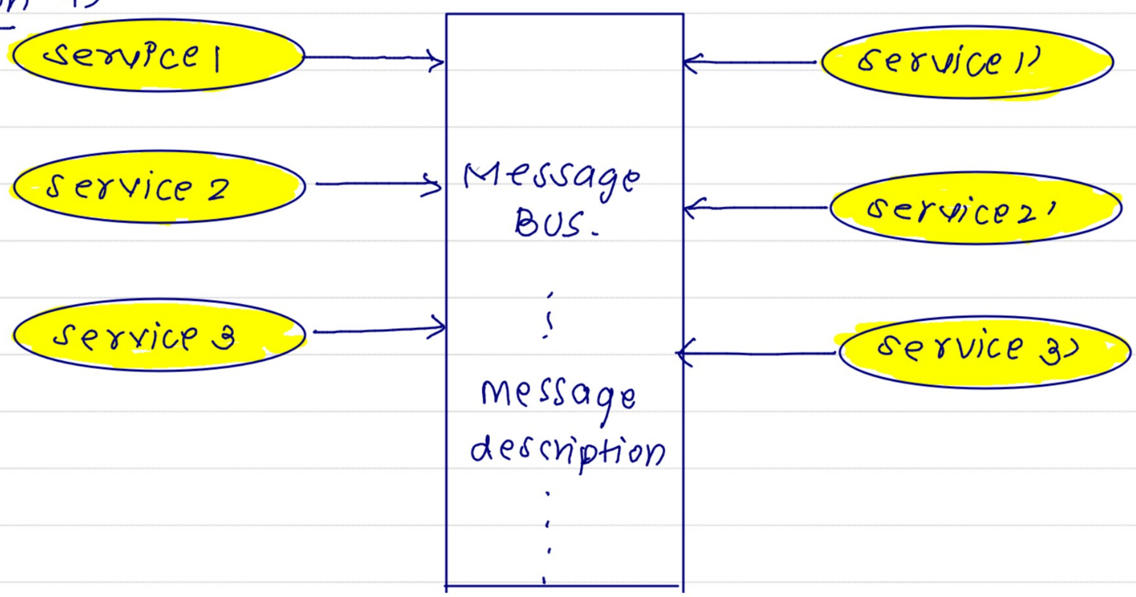
solution 9)



Message BUS.

solution a) A message bus is designed in way, where it can carry multiple services data into a single message bus to deliver to the releavant destination

ii) Many services align together, in a single message bus which in turn helps reduce dependencies between services in the system.

iii) In our given analogy multiple sensors and satellite services are invoked into a single message bus to be delivered to analyze pollution levels and alert to citizens

A b) I would choose topic-based filtering method for sending pollution alerts. This is because, the subscribed user is focused on only alerts here.

i) The conclusion of all the pollution monitoring services, would result into a set of only Alerts (High), No alert (Low)

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iii) Therefore I would use topic-based filtering here.

c) Message bus and Publish-subscribe model which helps to deliver alerts help in giving event driven alerts tha help city authorities to take action

i) In this providers; with help of "meta" data help only the subscribed users to receive notification

iii) once a pollution alert is triggered, subscribed users can then inform ~~authorities~~, to take subsequent steps to reduce pollution or reduce PM2.5 levels

iv) Therefore event-driven alerts can help city authorities to take immediate action

Q.No : 3 B) **Score : 3.00 / 5.00**

A research lab runs a complex climate simulation workflow on a distributed grid system. The workflow includes sequential preprocessing tasks and parallel computation tasks that require frequent data sharing across nodes.

a) Analyze how a DAG-based workflow architecture ensures correct task execution and efficient resource usage for this simulation.

b) Explain how the workflow execution engine handles control flow vs. data flow and ensures fault tolerance during long-running scientific computations.

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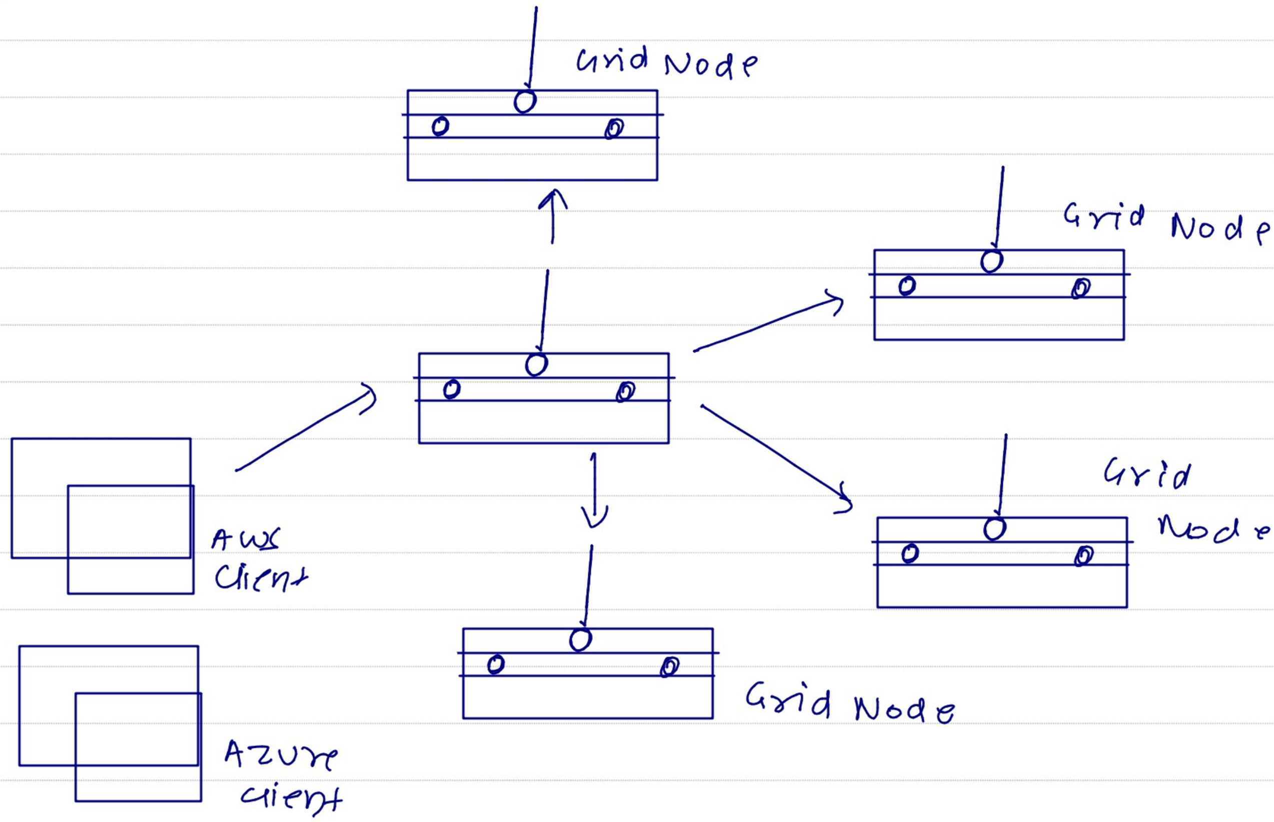
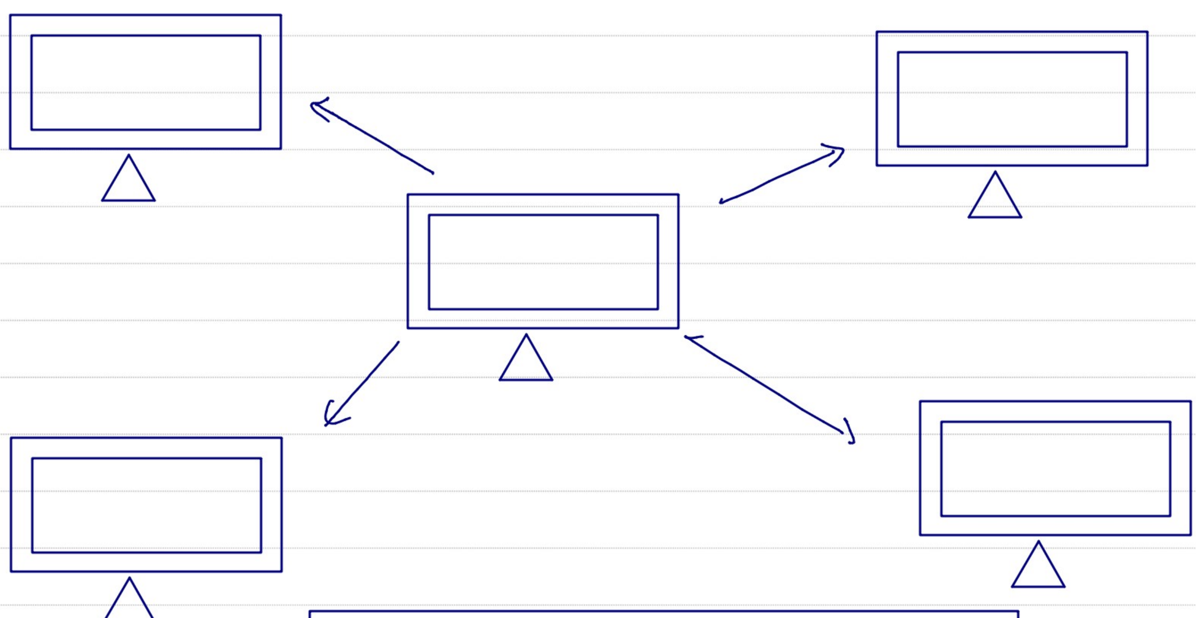




DISTRIBUTED GRID COMPUTING

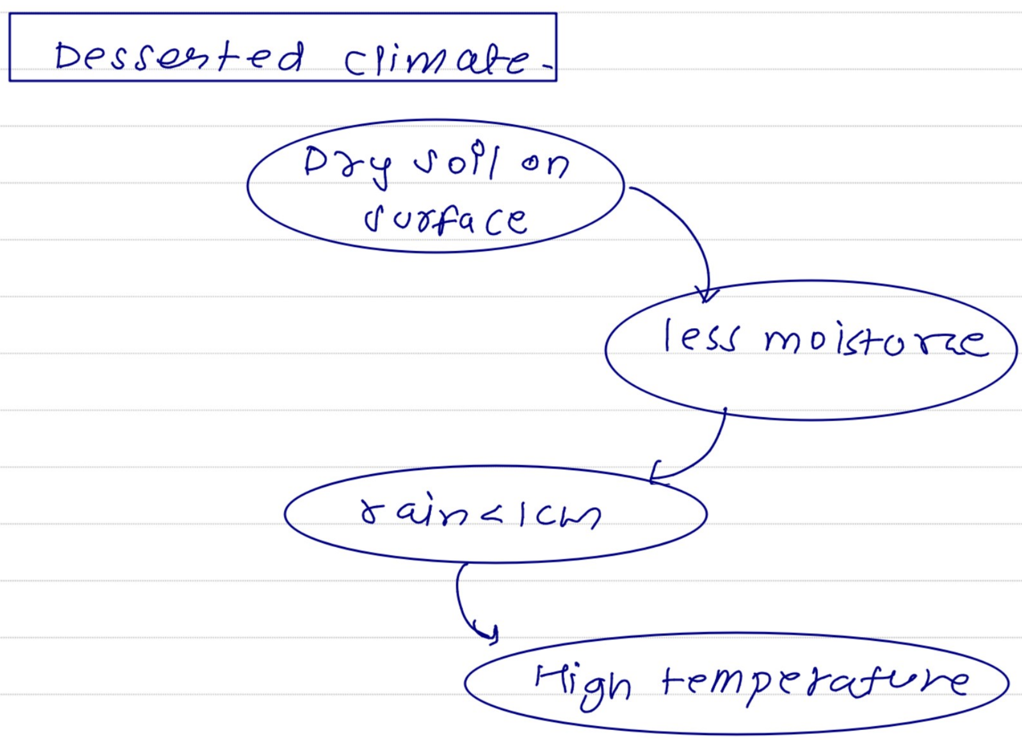
solution :- climate simulation workflow.
a) A DAG-based workflow architecture
ensures correct task execution and efficient

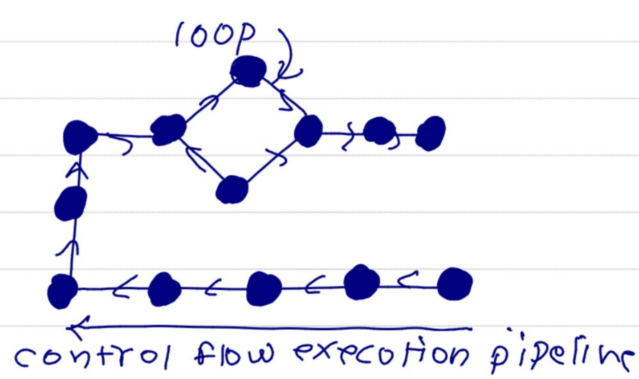
resource usage for this simulation.

(i) DAG (Directed Acyclic Graph) provides a workflow for synchronized task execution. and efficient ~~resource~~ simulation

(ii) For example let us consider a workflow of a desserted climate simulation

A DAG architecture of the same would be represented in





Q.No : 4 A)	**Score : 5.00 / 5.00**

A startup is developing a real-time food delivery tracking system. The application requires automatic scaling during peak meal times, fast deployment, and event-driven functions for processing live location updates.

a) Analyze how Google App Engine's Standard vs Flexible environments differ in supporting this application's scaling, runtime control, and execution requirements.

b) Evaluate whether AWS (Lambda, Elastic Beanstalk) or Microsoft Azure (Functions, App Services) provides a better programming environment for event-driven processing and rapid integration with backend storage - justify with service features.

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Solution:- Google App Engine is a software that is used for constructing web applications, that are fast, scalable and efficient in service oriented architectores (SOAs)

Scaling:- Google App Engine can help web apps in the field of food delivery" scale significantly.
ii) Google App Engine has enhanced inbuilt algorithms that helps in production level scaling during peak meal times.

runtime control: Google's App Engine can also help in higher runtimes and low latency especially when it comes to the up time in peak meal times

execution Environment: Google's App Engine uses execution environment such as visual studio, and visual studio code (VSC), improving robustness and easy for optimized processing for live location updates

Parameters	Application scaling	runtime control	execution requirements
Google App Engine	optimized	Low latency	visual studio visual studio code
Flexible Environment	depends on demand	High latency	Any.

b) According to me AWS (Lambda, Elastic Beenstalk) provides a better programming

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environment when it comes to food delivery tracking system. This is because AWS (Amazon web services) is provides highly fast and optimized processing of services, like ordering and payment, and its also very agile.

ii) AWS (Elastic Beanstalk) provides a scope of rapid integration of different services with a good backent storage facility.

iii) Backend retrival of data and database in AWS is faster when it comes to food delivery tracking system.

Q.No : 4 B) **Score : 4.50 / 5.00**

A cloud provider offers three kinds of agreements: one for individual users, one for enterprise customers, and one between internal departments of the same organization. Analyze your understanding of Service Level Agreements (SLAs) to identify the type of SLA represented in each case and explain how their scope and application differ in real-world cloud service scenarios.

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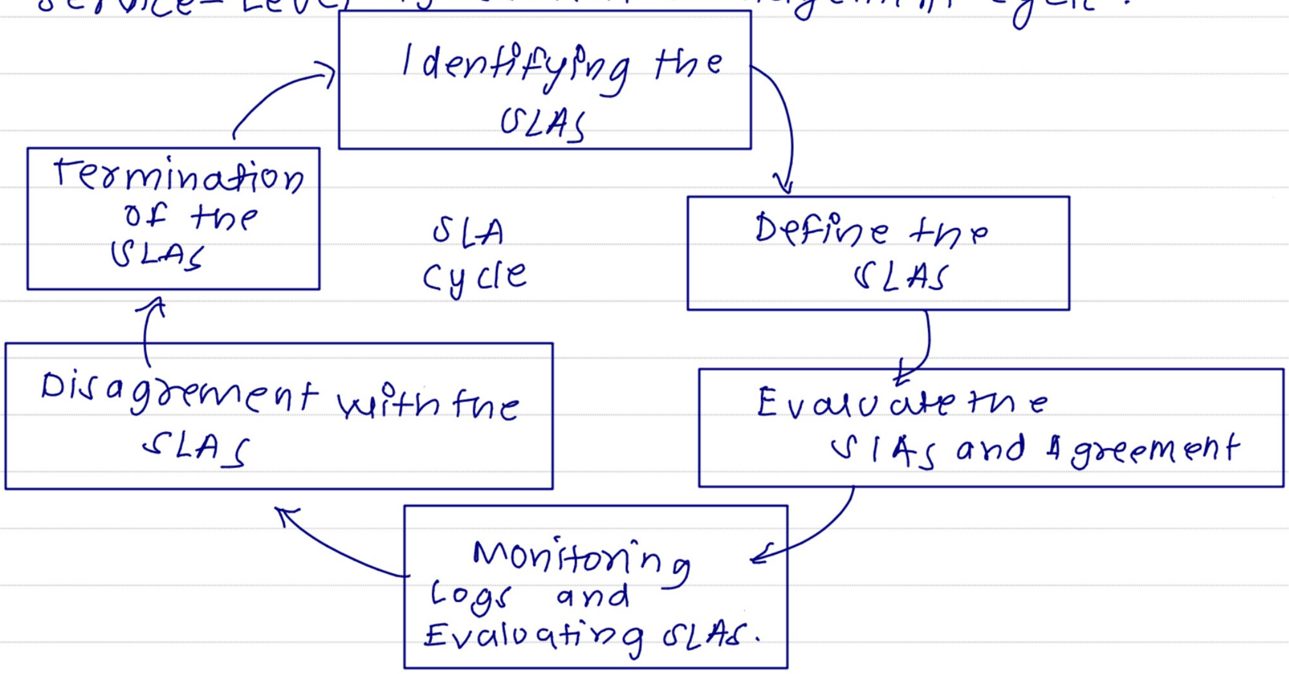
service-Level Agreement Management cycle.



solution:-

individual users Agreement :-

(i) An agreement is signed between the enterprise customers and individual users of the organization in this scenario.

ii) This agreement is a combined combination of two different services to work together in the execution of a application.

(ii) Each user recieves this agreement in the organisation, as a proof or authority to run a specific service

internal department agreement of same organization

i) This is also signed between the cloud provider, as a Evidence of services agreement between both the users of the organization and also the enterprise customers

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Enterprise customers:- i) Enterprise customers sign with the users that are present in the organization, in order to run a application by combining two services
ii) This helps the execution environment running until there is a disagreement between the enterprise customers and individual users of the organization.

Real-Life scenario:- we have An application "zepto", a service level "agreement (SLA) is passed between google maps and BHIM UPI services to integrate for the execution of the operations of the "zepto" App to deliver groceries on time.

Q.No : 5 A) **Score : 4.00 / 5.00**

Apply your knowledge of cloud security mechanisms to explain how encryption, strong authentication, and access control work together to protect sensitive customer data stored on a public cloud platform.

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Solution:- Encryption, strong authentication and access control work together to protect sensitive customer data storen on a public cloud:-

Strong authentication :- To protect strong sensitive customer data, we require authentication.

ii) It means, that we should know who is accessing the cloud, and prevent unauthorized users from modifying data in the public cloud

iii) Any unauthorized user can modify information of customer data, which can lead to serious consequences.

iv) For example. If we take an example of meta. cloud, unauthorized users can breach meta public cloud to steal data of customers and sell it in the black market.

v) Hence strong authentication is required to protect customer sensitive data.

Encryption : Data Encryption is another key mechanism necessary for protective sensitive customer data

ii) Encryption allows us to protect the data and store it in a hashed format, allowing enhanced privacy of sensitive information of customer data.

iii) For example, consider we have a cloud of Digilocker. The customer data like Aadhar card, pan card details, should be encrypted. In order to maintain customer detail privacy, if not taken care of hackers can use it to

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to take loans or destroy credit score of a user.

Access control : The Access control mechanism is a key mechanism, which is defined as who control's what. or simply authorization.

ii) In a scenario where our purpose is to protect sensitive customer data, we can assign different domains with relavant authorization.

iii) For example if we consider Digilocker, users should be authorized to view only their documents. Any other person cannot see a speific users' sensitive data, as it can be consequent of misuse of information if done so

iv) Therefore Access control of Authorization is extremly important to protect sensitive customer data on a public cloud platform.

Q.No : 5 B) **Score : 3.50 / 5.00**

A financial company uses a multi-layer cloud setup that includes firewalls, encryption, intrusion detection systems, and secure APIs to protect data across different service layers.

Explain how this setup reflects the concept of cloud computing security architecture and how layered protection enhances overall security.

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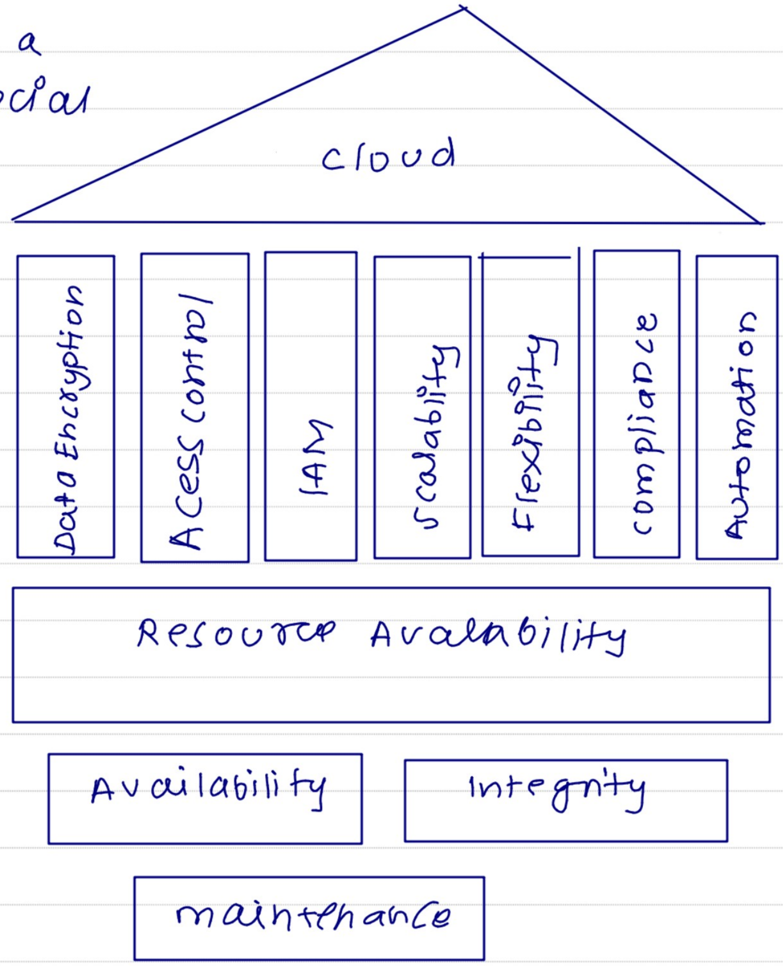
solution:- Let us consider a analogy where a Financial Company uses a multi-layer cloud setup that includes
(i) Firewalls
ii) Encryption
iii) intrusion detection systems
iv) secure APIs
to protect data across different service layers.



Firewalls: Firewalls mechanism play a crutial role in securing data in a multi-layer cloud setup of the Financial company
ii) Firewalls helps in detecting viruses and helps in preventing malware, and infected applications to function in a given space.
iii) Firewalls dads op as a Layer of security which prevent viruses and malware to prevent loss of data, theft / modify data in financial companies.

Encryption.: Encryption also prays a key role in adding another level of security to the cloud.
(i) Sensitive information included in the financial company, such as buisness model, customer

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account details, passwords must be secured in the cloud with Encrypted hash text.

ii) This is because, Even if a hacker achieves to breach from the firewall and has access to data, it would be encrypted in the form of hashed keys, which would be more difficult for the hacker to hack into the data.

iii) Hence encryption is necessary in the financial company.

Instrusion detection systems: Instrusion detection systems also adds up another layer of security to our multi-layer model.

ii) These systems identify data breaches and blocks them immediately

iii) These are important to maintain data intergrity and security.

Secure API: Secure APIs (Application Programming Interfaces) like payment API should be present in order of secure data transfer of funds from a company to Buisness.

In real life scenario, this enables the secure transfer of sensitive data, without breaching.